

Smooth Sensorless Startup in PMSM Drives Through Cogging Torque Mitigation

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Cogging torque, an inherent pulsating torque in permanent magnet machines, can induce position and speed fluctuations during startup, potentially leading to inaccuracies in position estimation for sensorless control. Improving estimation precision and minimizing speed fluctuations during startup necessitate the mitigation of cogging torque in sensorless control. To address this challenge, this article proposes a new signal injection strategy. The key of the proposed method is to inject a high-frequency signal into the d-axis for accurate position estimation, and a low-frequency signal into the q-axis to counteract the cogging torque. Through experimental validation, the proposed method has exhibited significant improvements in position estimation accuracy and promoted smoother torque characteristics.

Index Terms—Permanent magnet machine, position sensorless control, torque.

I. INTRODUCTION

THE permanent magnet synchronous machine (PMSM) with field-orientation control (FOC) drives has attracted significant attention due to its notable advantages, including high power density and superior control performance. However, the precise rotor position is crucial for FOC drives, necessitating the use of a position sensor. With advancements in research, position sensorless control has emerged as a prominent focus, offering benefits like cost efficiency, compact structure, and redundancy in case of sensor failures, leading to its widespread adoption in industrial drive systems [1].

At present, estimation during the startup phase relies on the rotor saliency feature of the PMSM and is commonly accomplished through signal injection [2]. While existing research has explored various aspects of injection-based sensorless control, the issue of cogging torque during the startup phase of PMSMs remains inadequately examined.

Cogging torque T_{cog} is distinct for its persistent presence and noticeable impact throughout the operational range of PMSMs, particularly during machine startup and under low-load conditions. This presents significant challenges for achieving high-performance sensorless control.

Therefore, mitigating cogging torque is essential for enhancing the performance of sensorless applications. While design approaches can aid in reducing cogging torque [3], many factors post-production can worsen cogging torque, such as manufacturing imperfections, assembly inaccuracies, rotor shaft misalignment, and demagnetization of the PMs. These issues can damage the PMSM startup performance.

To mitigate the effects of cogging torque during startup sensorless control, a mixed-signal injection method is proposed for the first time in this article.

II. PROPOSED SMOOTH SENSORLESS CONTROL METHOD

Overall, the block diagram of the proposed control strategy is illustrated in Fig. 1. Two sinusoidal signals are injected into the dq-axes: a low-frequency (LF) signal is injected to the q-axis to realize cogging torque mitigation (CTM), while a high-

frequency (HF) signal is injected into the d-axis. Subsequently, the rotor position can be estimated through the analysis of the reflected winding currents. The injection of two non-interfering signals ensures the achievement of both control objectives.

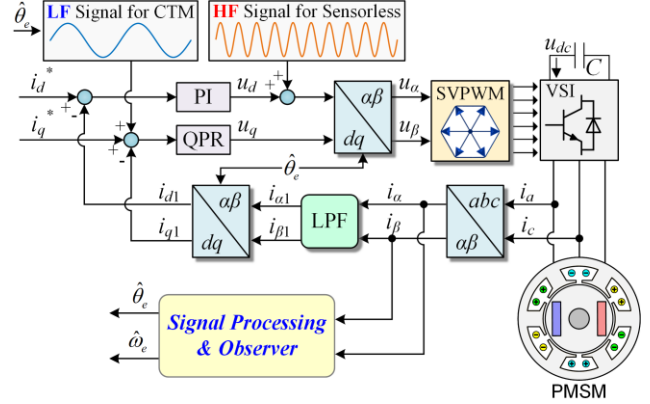


Fig. 1. Block diagram of the proposed control strategy.

A. Principle of Injection-Based Sensorless Control

A high-frequency sinusoidal voltage signal is injected into the $\hat{d}$ -axis of the estimated synchronous rotating coordinate frame, followed by detection of the corresponding current response generated in the motor.

In the estimated rotor synchronous rotating reference frame, the $\hat{q}$ -axis high-frequency current component is related to the rotor position estimation error angle. When the rotor position estimation error angle is zero, the $\hat{q}$ -axis high-frequency current becomes zero. Consequently, rotor position information can be extracted through a specific signal processing procedure [4].

B. Principle of Injection-Based CTM

After the initial position estimation, the estimated position converges to the actual rotor position and is employed to calculate the harmonic signal.

Harmonic torque, T_{eh} , is generated by current harmonics and adheres to the classical electromagnetic torque equation [5]. Therefore, by appropriately selecting the q-axis harmonic current, a balance between the two torque components can be achieved, such that $T_{\text{eh}} = -T_{\text{cog}}$, thereby eliminating the adverse effect of cogging torque during sensorless startup.

C. Mutual Influence Analysis of Injection Signals

The derivation and simulation results both demonstrate that independent injection into the d- and q-axes eliminates signal coupling, thereby enabling simultaneous sensorless control and low torque pulsation startup. Fig. 2 presents the current and torque results of different winding excitation methods. It is evident that the LF i_{qh} for CTM is highly effective, and the HF i_{dh} does not lead to significant additional torque ripple.

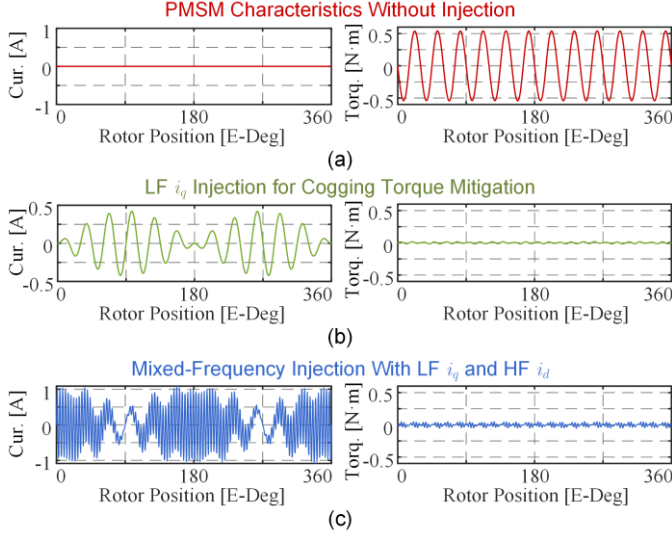


Fig. 2. Phase-A current and torque waveforms under different winding excitation methods based on FEM simulations. (a) Without injection. (b) LF injection. (c) Mixed-frequency injection.

III. EXPERIMENTAL VERIFICATION

Experimental validation was conducted to verify the effectiveness of the proposed control strategy and the correctness of the theoretical analysis.

A. Sensorless Startup Control

First, the motor startup control experiment was conducted using both the conventional HF d-axis voltage injection method and the proposed method. The experiment recorded the actual position, estimated position, position estimation error, and shaft torque. The experimental waveforms are shown in Fig. 3. It can be observed that the torque ripple of the proposed method is significantly lower than that of the conventional method.

B. Sensorless Control Under On-Load Conditions

Secondly, on-load experiments under sensorless control were conducted using both the conventional and proposed methods. The motor was operated at a mechanical speed of 90 rpm and a rated torque of 5 N·m. The experimental waveforms are shown in Fig. 4. It can be observed that, under load conditions, the proposed method remains effective in suppressing cogging torque and results in smaller shaft torque ripple. This leads to smoother torque and power output and reduced shaft fatigue.

Moreover, the reduced torque ripple enables sensorless control to operate under lower signal-to-noise ratio conditions, as the cogging torque disturbance is minimized. This allows for the use of a smaller injection voltage or lower injection frequency, thereby reducing associated losses and noise.

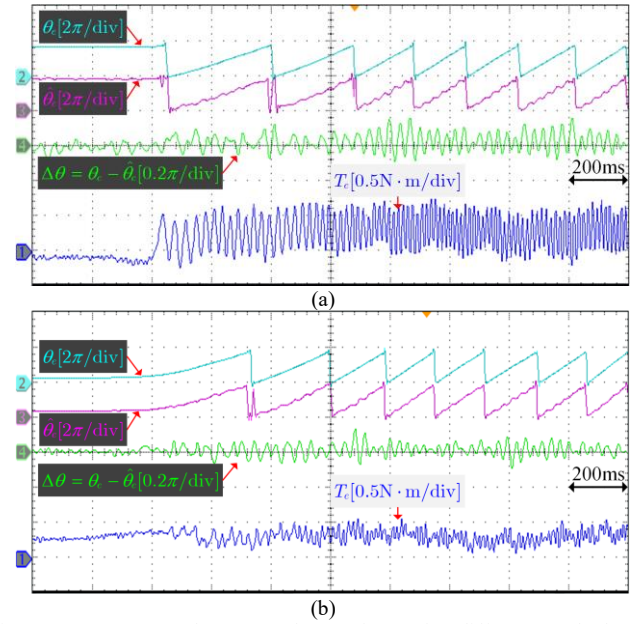


Fig. 3. Startup sensorless control waveforms for different methods. (a) Conventional method. (b) Proposed method.

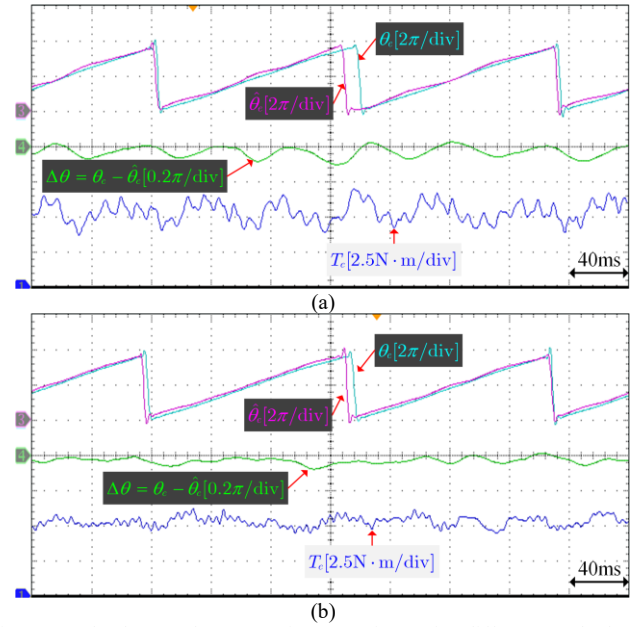


Fig. 4. On-load sensorless operation waveforms for different methods. (a) Conventional method. (b) Proposed method.

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